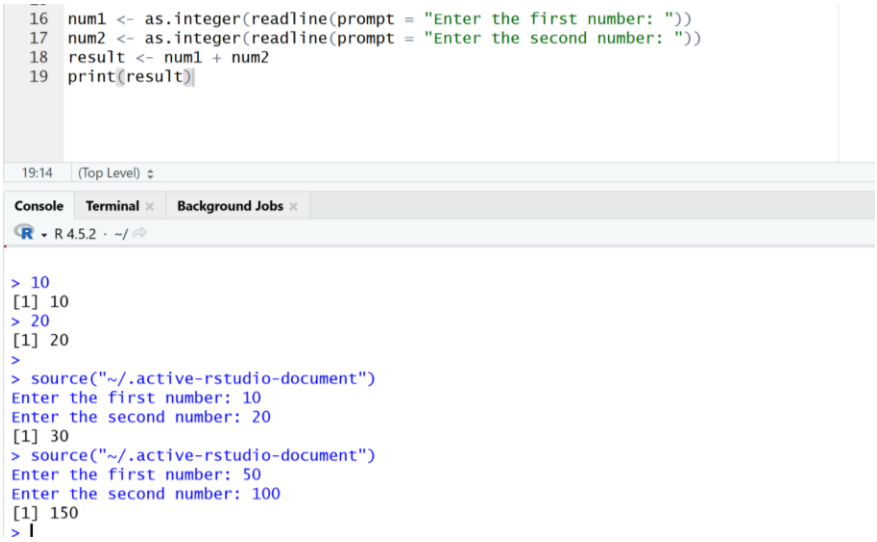
Lab Experiments
CSA-1614-Data warehousing and Data minning

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experiment 1



```
num1 <- as.integer(readline(prompt = "Enter the first number: "))
num2 <- as.integer(readline(prompt = "Enter the second number: "))
result <- num1 + num2
print(result)
```

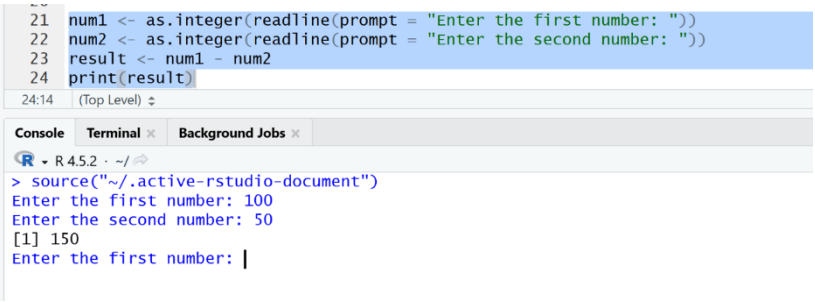
19:14 (Top Level)

Console Terminal Background Jobs

R 4.5.2 · ~/

```
> 10
[1] 10
> 20
[1] 20
>
> source("~/.active-rstudio-document")
Enter the first number: 10
Enter the second number: 20
[1] 30
> source("~/.active-rstudio-document")
Enter the first number: 50
Enter the second number: 100
[1] 150
>
```

experiment 2



```
num1 <- as.integer(readline(prompt = "Enter the first number: "))
num2 <- as.integer(readline(prompt = "Enter the second number: "))
result <- num1 - num2
print(result)
```

24:14 (Top Level)

Console Terminal Background Jobs

R 4.5.2 · ~/

```
> source("~/.active-rstudio-document")
Enter the first number: 100
Enter the second number: 50
[1] 150
Enter the first number:
```

experiment 3

```
3.
num1 <- as.integer(readline(prompt = "Enter the first number: "))
num2 <- as.integer(readline(prompt = "Enter the second number: "))
result <- num1 * num2
print(result)
```

```
> source("~/.active-rstudio-document")
Enter the first number: 5
Enter the second number: 10
[1] 50
>
```

experiment 4

```
num1 <- as.numeric(readline(prompt = "Enter the first number: "))
num2 <- as.numeric(readline(prompt = "Enter the second number: "))

if (num2 != 0) {
  result <- num1 / num2
  print(result)
} else {
  print("Cannot divide by zero")
}
```

```
> source("~/.active-rstudio-document")
Enter the first number: 10
Enter the second number: 2
[1] 5
>
```

experiment 5

```
num <- as.integer(readline(prompt = "Enter a number: "))

if (num %% 2 == 0) {
  print("Number is even")
} else {
  print("Number is odd")
}
```

```
> source("~/.active-rstudio-document")
Enter a number: 23
[1] "Number is odd"
>
```

experiment 6

```r
student_data <- data.frame(
  name = c("Arun", "Priya", "Kiran", "Meena", "Ravi"),
  age = c(21, 22, 22, 24, 25),
  marks = c(78, 85, 91, 85, 88)
)

find_mode <- function(x) {
  values <- unique(x)
  values[which.max(tabulate(match(x, values)))]
}

print(mean(student_data$age))
print(median(student_data$age))
print(find_mode(student_data$age))
```

```
> student_data <- data.frame(
+   name = c("Arun", "Priya", "Kiran", "Meena", "Ravi"),
+   age = c(21, 22, 22, 24, 25),
+   marks = c(78, 85, 91, 85, 88)
+ )
>
> find_mode <- function(x) {
+   values <- unique(x)
+   values[which.max(tabulate(match(x, values)))]
+ }
>
> print(mean(student_data$age))
[1] 22.8
> print(median(student_data$age))
[1] 22
> print(find_mode(student_data$age))
[1] 22
>
> write.csv(student_data, "student_data.csv", row.names = FALSE)
```

experiment 7

```r
students <- data.frame(
  name = c("Anu", "Bala", "Charan", "Divya", "Rohan"),
  age = c(19, 20, 21, 22, 23),
  marks = c(76, 84, 90, 68, 95)
)

result <- summary(students)
print(result)

write.csv(students, "student_summary.csv", row.names = FALSE)

write.csv(student_data, "student_data.csv", row.names = FALSE)
```

```
> students <- data.frame(
+   name = c("Anu", "Bala", "Charan", "Divya", "Rohan"),
+   age = c(19, 20, 21, 22, 23),
+   marks = c(76, 84, 90, 68, 95)
+ )
>
> result <- summary(students)
> print(result)
     name                age        marks
 Length:5           Min.   :19   Min.   :68.0
 Class :character   1st Qu.:20   1st Qu.:76.0
 Mode  :character   Median :21   Median :84.0
                    Mean   :21   Mean   :82.6
                    3rd Qu.:22   3rd Qu.:90.0
                    Max.   :23   Max.   :95.0
>
> write.csv(students, "student_summary.csv", row.names = FALSE)
>
> write.csv(student_data, "student_data.csv", row.names = FALSE)
```

experiment 8

```r
first <- as.numeric(readline("Enter the first value: "))
second <- as.numeric(readline("Enter the second value: "))
third <- as.numeric(readline("Enter the third value: "))

values <- c(first, second, third)
greatest <- max(values)

cat("The greatest number is:", greatest, "\n")
```

```
> source("~/.active-rstudio-document")
Enter the first value: 980
Enter the second value: 520
Enter the third value: 976
The greatest number is: 980
>
```

experiment 9

```r
student_data <- data.frame(
  name = c("Arun", "Priya", "Kiran", "Meena", "Ravi"),
  age = c(20, 21, 22, 23, 24),
  marks = c(55, 67, 78, 89, 95)
)

result <- IQR(student_data$marks)
print(result)

write.csv(student_data, "iqr_data.csv", row.names = FALSE)
```

```
> student_data <- data.frame(
+   name = c("Arun", "Priya", "Kiran", "Meena", "Ravi"),
+   age = c(20, 21, 22, 23, 24),
+   marks = c(55, 67, 78, 89, 95)
+ )
>
> result <- IQR(student_data$marks)
> print(result)
[1] 22
>
> write.csv(student_data, "iqr_data.csv", row.names = FALSE)
```

experiment 10

```r
student_data <- data.frame(
  name = c("Asha", "Bala", "Charan", "Divya", "Rohan"),
  age = c(19, 20, 21, 22, 23),
  marks = c(65, 72, 81, 88, 96)
)

result <- quantile(student_data$marks, probs = c(0, 0.25, 0.5, 0.75, 1))
print(result)

write.csv(student_data, "quantile_data.csv", row.names = FALSE)
```

```
> student_data <- data.frame(
+   name = c("Asha", "Bala", "Charan", "Divya", "Rohan"),
+   age = c(19, 20, 21, 22, 23),
+   marks = c(65, 72, 81, 88, 96)
+ )
>
> result <- quantile(student_data$marks, probs = c(0, 0.25, 0.5, 0.75, 1))
> print(result)
  0%  25%  50%  75% 100%
  65   72   81   88   96
>
> write.csv(student_data, "quantile_data.csv", row.names = FALSE)
```

experiment 11

```r
scores <- c(45, 60, 72, 85, 95)

minimum <- min(scores)
maximum <- max(scores)
mid_range <- (minimum + maximum) / 2

print(minimum)
print(maximum)
print(mid_range)
```

```
> scores <- c(45, 60, 72, 85, 95)
>
> minimum <- min(scores)
> maximum <- max(scores)
> mid_range <- (minimum + maximum) / 2
>
> print(minimum)
[1] 45
> print(maximum)
[1] 95
> print(mid_range)
[1] 70
>
```

experiment 12

```r
marks <- c(55, 65, 70, 80, 90, 95)

average <- mean(marks)
std_dev <- sd(marks)

z_scores <- (marks - average) / std_dev

print(z_scores)
```

```
> marks <- c(55, 65, 70, 80, 90, 95)
>
> average <- mean(marks)
> std_dev <- sd(marks)
>
> z_scores <- (marks - average) / std_dev
>
> print(z_scores)
[1] -1.3614329 -0.7079451 -0.3812012  0.2722866  0.9257743  1.2525182
>
```

13. experiment 13

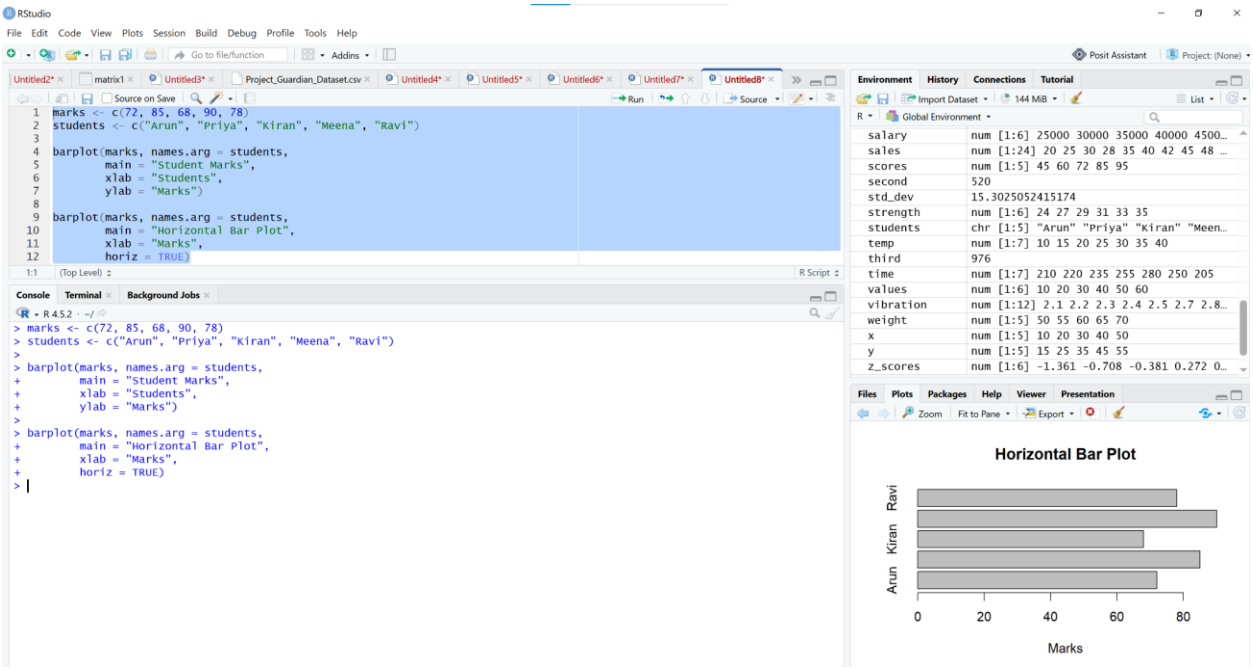
```r
values <- c(10, 20, 30, 40, 50, 60)
average <- mean(values)
minimum <- min(values)
maximum <- max(values)

normalized <- (values - minimum) / (maximum - minimum)

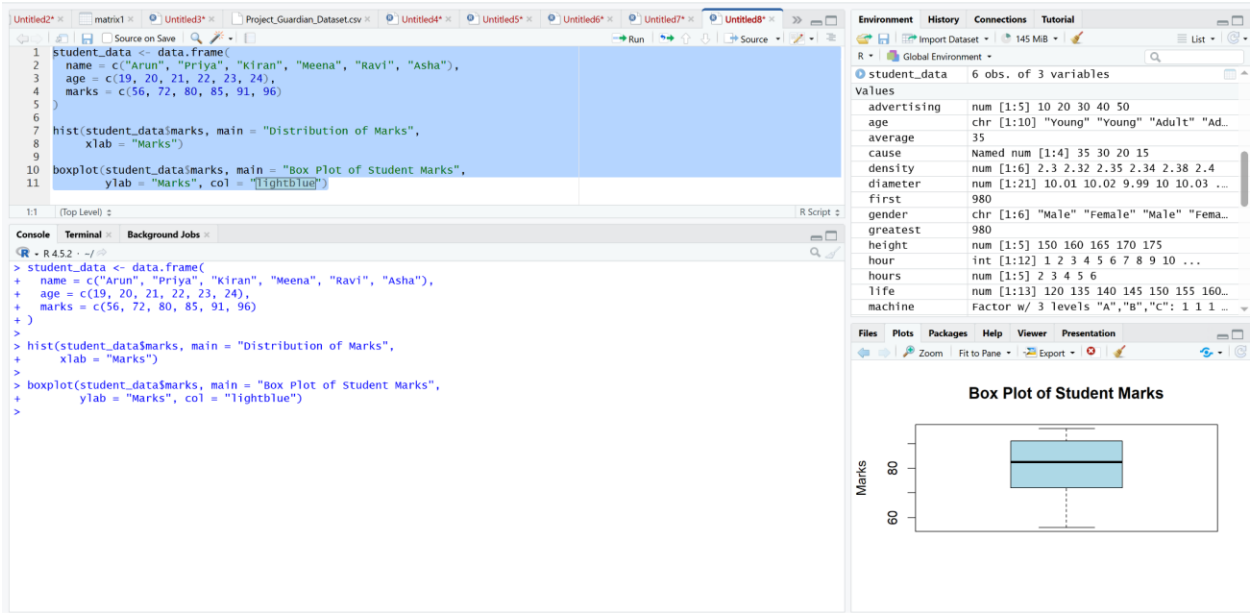
print(average)
print(minimum)
print(maximum)
print(normalized)
```

```
> values <- c(10, 20, 30, 40, 50, 60)
> average <- mean(values)
> minimum <- min(values)
> maximum <- max(values)
>
> normalized <- (values - minimum) / (maximum - minimum)
>
> print(average)
[1] 35
> print(minimum)
[1] 10
> print(maximum)
[1] 60
> print(normalized)
[1] 0.0 0.2 0.4 0.6 0.8 1.0
```

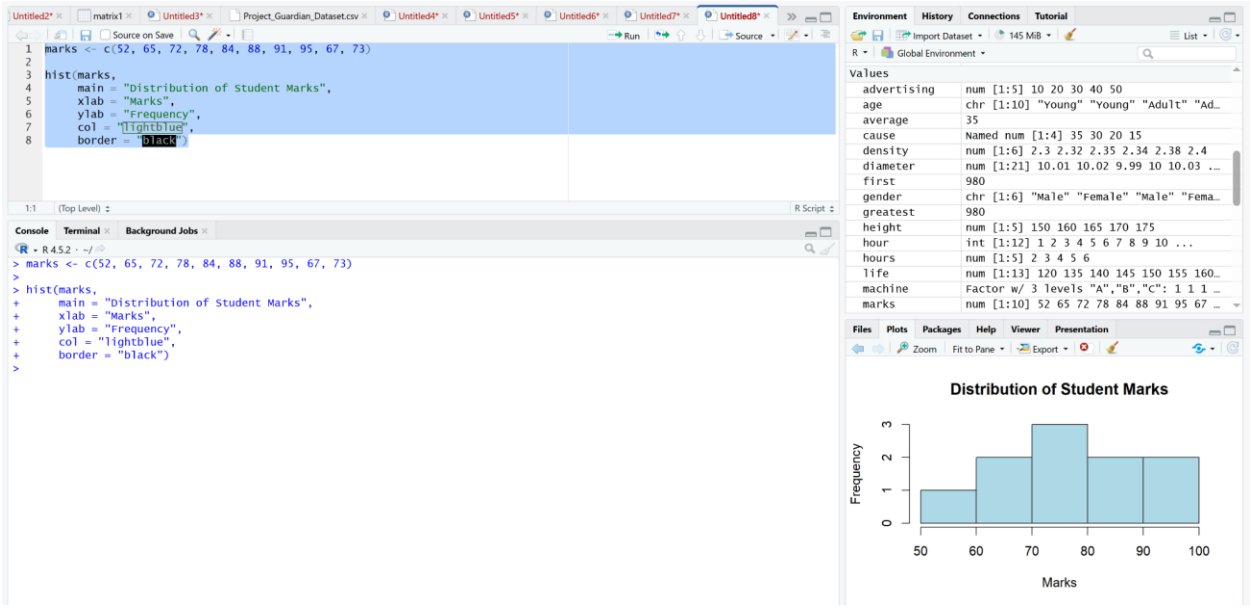
14. experiment 14



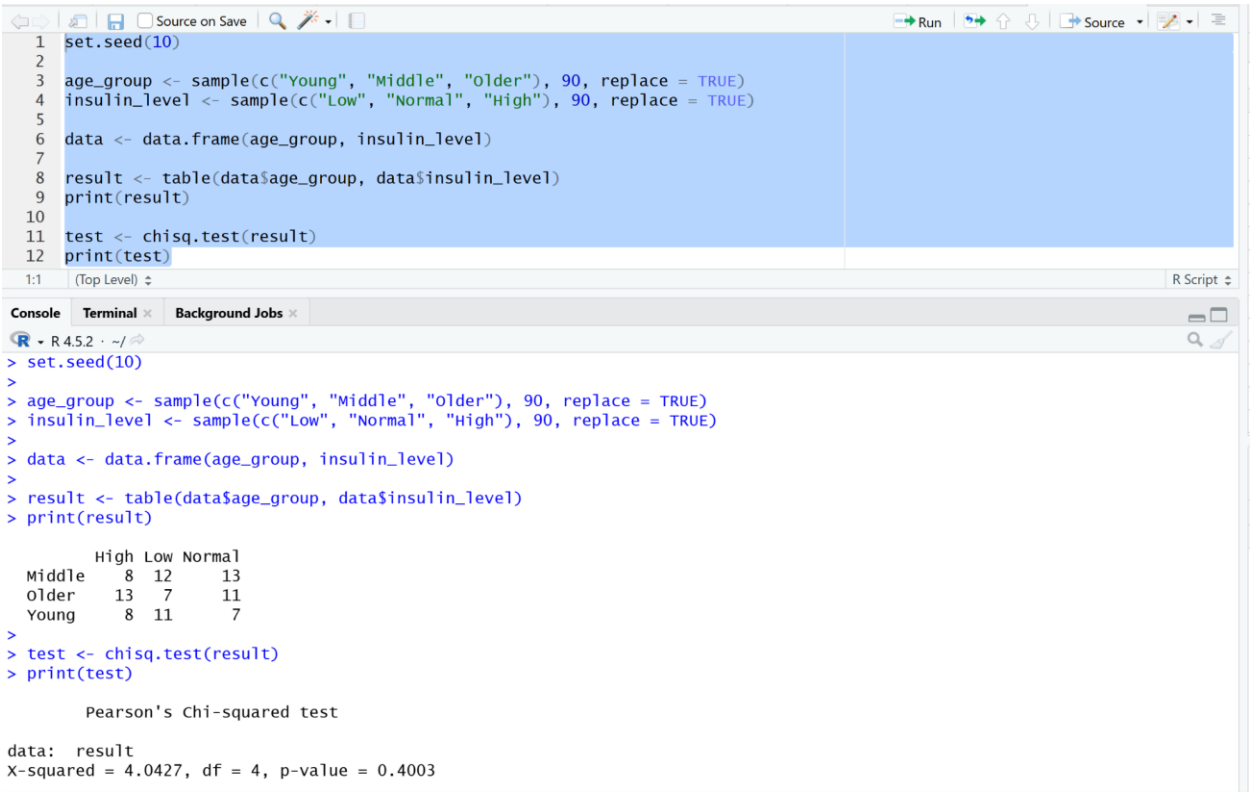
15. experiment 15



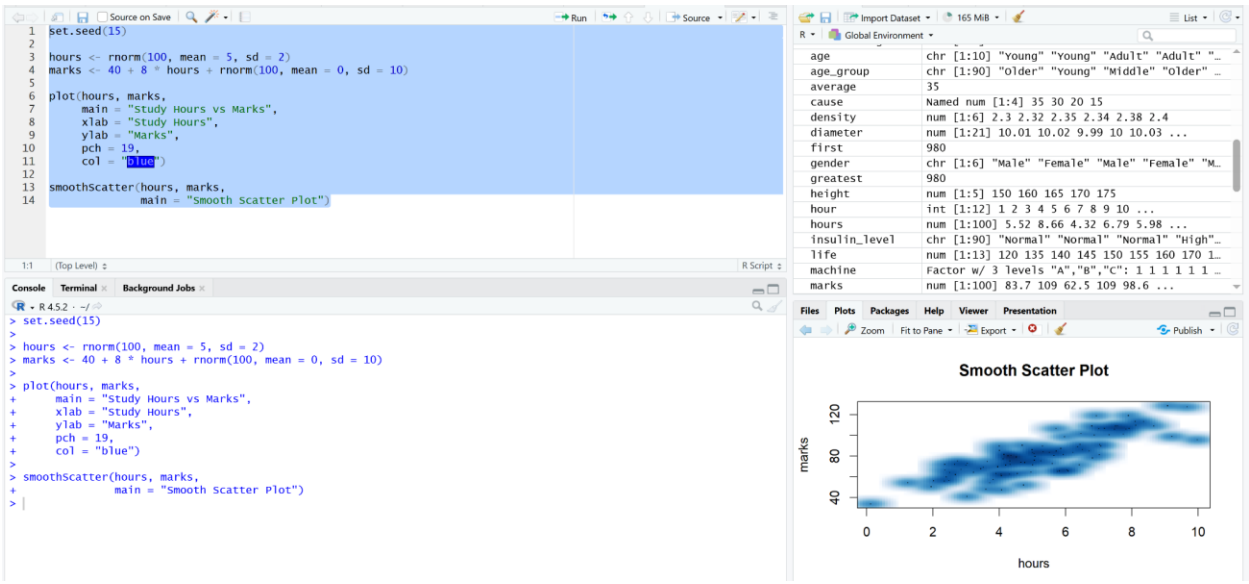
16. experiment 16



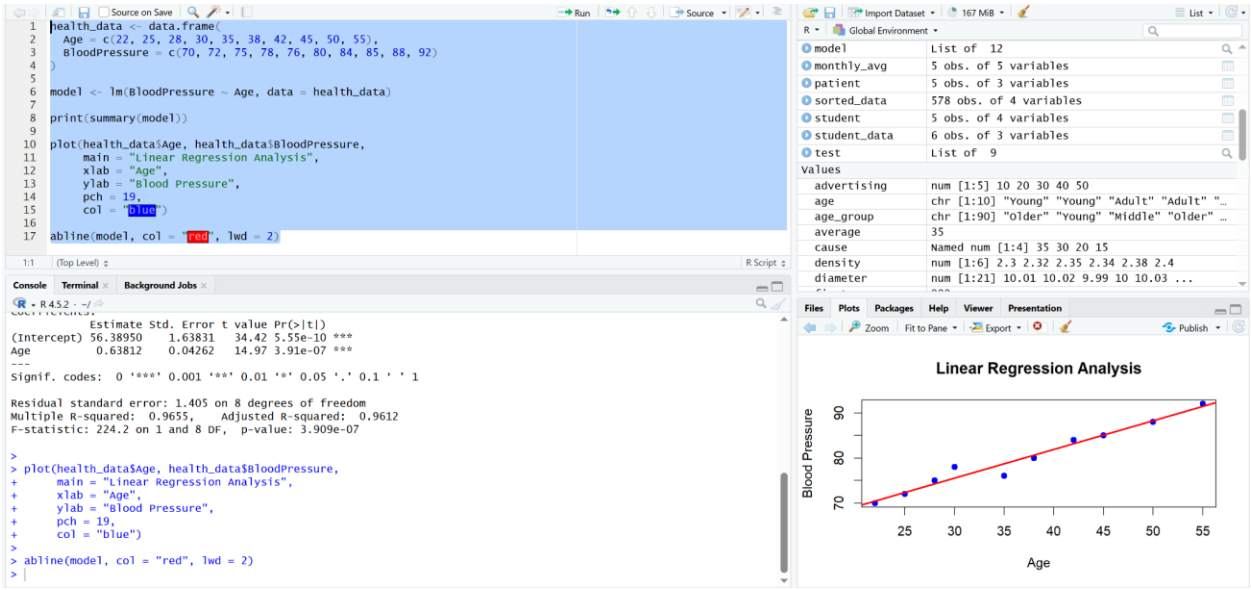
17. experiment 17



18. experiment 18



19. experiment 19



20. experiment 20

```r
  Age = c(22, 25, 28, 30, 35, 38, 42, 45, 50, 55),
  BloodPressure = c(70, 72, 75, 78, 76, 80, 84, 85, 88, 92),
  Glucose = c(90, 95, 100, 105, 110, 115, 120, 125, 130, 140)
)

model <- lm(Age ~ BloodPressure + Glucose, data = health_data)

print(model)
print(coef(model))
```

1:1 (Top Level) R Script

Console Terminal Background Jobs

R 4.5.2 · ~/

```
Call:
lm(formula = Age ~ BloodPressure + Glucose, data = health_data)

Coefficients:
  (Intercept)  BloodPressure        Glucose
     -37.1519        -0.1034         0.7294

> print(coef(model))
  (Intercept) BloodPressure       Glucose
  -37.1518645    -0.1034048     0.7294181
> print(summary(model))

Call:
lm(formula = Age ~ BloodPressure + Glucose, data = health_data)

Residuals:
    Min      1Q  Median      3Q     Max
-1.3715 -0.3991 -0.1300  0.3063  1.4271

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   -37.1519     7.8024  -4.762 0.002056 **
BloodPressure  -0.1034     0.2481  -0.417 0.689370
Glucose         0.7294     0.1105   6.602 0.000304 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.8601 on 7 degrees of freedom
Multiple R-squared:  0.9952,    Adjusted R-squared:  0.9939
F-statistic: 730.4 on 2 and 7 DF,  p-value: 7.489e-09
```